

The Ultimate Question Analysis Guide

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1 The Ultimate Question Analysis Guide

What this is: A routing guide. Every question type you might face is mapped to the exact method, R command, and cheatsheet section that solves it.

How to use it: 1. Read the question once. 2. Find the **verb** (Part 2) or the **keyword** (Part 1). 3. Jump to the topic decision tree. 4. Follow the branch until you reach a “do this” cell. 5. Open the corresponding cheatsheet section and execute. 6. Output in the shape shown in Part 12.

Rule: If you can find your question in this guide, you can solve it. If you can’t, the question is out of scope.

2 Part 0 — The Meta-Strategy

2.1 0.1 The 4-step reading protocol

For **every** question, do these four things before writing any code:

1. **Underline the verb.** What is the question asking you to *do*? (Fit? Test? Estimate? Compare? Forecast? Interpret?)
2. **Circle the nouns.** What are the variables? Which is the response? Which are predictors? How many? What types (continuous, categorical, count, binary, time-indexed)?
3. **Identify the design.** One sample? Multiple groups? Repeated measures? Time series? Two variables? Many variables?
4. **Pick the output shape.** Point estimate? Interval? Test statistic? Model object? Plot? Two numbers? Table?

Then open the relevant decision tree below.

2.2 0.2 The universal fallback

If you can’t classify the question immediately:

If the question has...	Then try...
One numeric response, one numeric predictor	Regression (Part 4)
One numeric response, one categorical predictor	ANOVA (Part 6)
One numeric response, multiple predictors	Regression (Part 4) or ANOVA (Part 6)
Two categorical variables	Categorical (Part 9)
Multiple numeric responses	Multivariate (Part 7)
A time index	Time series (Part 3)
A finite population + sample	Sampling (Part 8)
A data file to clean first	Data handling (Part 5)
A binary response	Categorical (Part 9) or Regression (Part 4, logistic)
A count response	Regression (Part 4, Poisson)

2.3 0.3 The five question shapes

Every exam question is one of these five shapes:

Shape	Typical verbs	Output
Describe	describe, discuss, plot, summarise	Words + plot + numbers

Shape	Typical verbs	Output
Estimate	estimate, compute, calculate	Point + SE + CI
Test	test, is there, does, compare	Statistic + df + p + conclusion
Model	fit, build, select, choose	Model object + coefficients
Predict	predict, forecast	Point + PI

If your question starts with a verb not in this table, it's a compound question. Break it into these five shapes.

2.4 0.4 When to stop and think

Stop before coding if any of these are true:

- The question has **more than 20 words** of setup.
- The question asks you to **explain why**.
- The question mentions **“assumptions”** or **“check”**.
- The question says **“compare”**.
- The question says **“interpret”**.

These are the questions where students lose marks by rushing to code.

3 Part 1 — Keyword Index

Search the question for any of these words. The right column tells you where to go.

3.1 1.1 Statistical keywords

Keyword in question	Topic	Section
Stationary, unit root, ADF	Time series	3
ACF, PACF, autocorrelation	Time series	3
Forecast, h steps ahead	Time series	3
Seasonal, period, frequency	Time series	3
ARIMA, AR, MA, SARIMA	Time series	3
GARCH, volatility, ARCH	Time series	3
Slope, intercept, R^2	Regression	4
Residuals, fitted values	Regression	4
Multicollinearity, VIF	Regression	4
Interaction, effect modification	Regression / ANOVA	4, 6
ANCOVA, covariate	Regression / ANOVA	4, 6
Box-Cox, transformation	Regression	4
Logistic, odds ratio, binary	Regression / Categorical	4, 9
Poisson, count, rate	Regression	4
Overdispersion	Regression	4
Ridge, lasso, penalized	Regression	4
Group means, treatment effect	ANOVA	6
Blocking, block, RCBD	ANOVA	6
Factor, factorial, level	ANOVA	6
Split-plot, whole-plot	ANOVA	6
Repeated measures, within-subject	ANOVA	6
Mixed model, random effect	ANOVA	6
Post-hoc, Tukey, Bonferroni	ANOVA	6

Keyword in question	Topic	Section
Effect size, eta-squared	ANOVA	6
Power, sample size	ANOVA / Sampling	6, 8
Randomization, replication	ANOVA	6
Latin Square, BIBD	ANOVA	6
Fractional factorial, resolution	ANOVA	6
Mean vector, MANOVA	Multivariate	7
Hotelling, Wilks, Pillai	Multivariate	7
Multivariate normality, Mardia	Multivariate	7
PCA, principal component	Multivariate	7
Factor analysis, latent, loadings	Multivariate	7
Discriminant, LDA, QDA	Multivariate	7
Cluster, k-means, dendrogram	Multivariate	7
Mahalanobis, multivariate outlier	Multivariate	7
Canonical correlation	Multivariate	7
SRS, simple random	Sampling	8
Stratified, stratum, strata	Sampling	8
Cluster sampling	Sampling	8
Systematic sampling	Sampling	8
FPC, finite population correction	Sampling	8
Bias, MSE, relative efficiency	Sampling	8
Coverage, confidence interval	Sampling	8
Ratio estimator, regression estimator	Sampling	8
PPS, size-biased	Sampling	8
Horvitz-Thompson	Sampling	8
Design effect, deff	Sampling	8
Contingency table, cross-tab	Categorical	9
Chi-square, independence	Categorical	9
Homogeneity, goodness-of-fit	Categorical	9
Fisher exact	Categorical	9
McNemar, paired binary	Categorical	9
Cochran's Q	Categorical	9
Cramér's V, phi	Categorical	9
Mantel-Haenszel	Categorical	9
Simpson's paradox	Categorical	9
Log-linear	Categorical	9
Clean, tidy, reshape, merge	Data handling	5
Missing, NA, impute	Data handling	5
Outliers	Data handling	5

3.2 1.2 Mathematical keywords

Keyword	Meaning	Action
"Derive"	Prove a formula	Show steps, use algebra
"Show that"	Prove a property	Start from definition, manipulate
"Verify"	Check numerically	Compute both sides
"Justify"	Explain the choice	Cite assumption or theorem
"Interpret"	Explain in context	Say what the number means for the data
"State"	Give without proof	One line, no justification
"Compute"	Numerical answer	Show formula, plug in, give number

Keyword	Meaning	Action
“Approximate”	Use CLT or asymptotics	Note the approximation

4 Part 2 — Question Shape Classifier

4.1 2.1 By verb

The question says...	You should...	Output
“Describe”	EDA: summary stats + plot	Paragraph + table
“Discuss”	EDA + interpretation	Paragraph
“Plot”	<code>ggplot</code> or base plot	Plot
“Summarise”	<code>group_by %>% summarise</code>	Table
“Estimate”	Point estimate + SE + CI	Number + interval
“Compute”	Exact calculation	Number
“Calculate”	Exact calculation	Number
“Test”	Hypothesis test	Statistic, df, p, conclusion
“Is there”	Hypothesis test	Same
“Does”	Hypothesis test	Same
“Compare”	Test of difference	Same + effect size
“Is it significant”	Hypothesis test	Same
“Fit”	Model fitting	Model object + coefficients
“Build”	Model fitting	Same
“Select” / “Choose”	Model selection	AIC/BIC or CV result
“Predict”	<code>predict()</code>	Point + interval
“Forecast”	<code>forecast()</code>	Point + PI
“Check”	Diagnostics	Plots + tests
“Assess”	Diagnostics	Same
“Interpret”	Translate coefficient/statistic	Sentence
“Explain”	Explain mechanism	Paragraph
“Why”	Explain mechanism	Paragraph
“Design”	Design an experiment	Design table
“Simulate”	Monte Carlo	Code + summary
“Derive”	Mathematical proof	Algebra
“Sketch”	Draw a rough plot	Hand-drawn or code
“State”	Give a fact	One-line answer

4.2 2.2 By question structure

Structure	Type
“Is X related to Y?”	Association test
“Do groups differ?”	Group comparison
“Does X affect Y?”	Effect estimation
“What is the effect of X on Y?”	Effect estimation
“Can we predict Y from X?”	Regression / prediction
“How many X do we need?”	Sample size
“What is the best model?”	Model selection
“Is the model adequate?”	Diagnostics
“What does this coefficient mean?”	Interpretation

Structure	Type
“Describe the relationship”	EDA + regression

4.3 2.3 By data shape

Data has...	Likely topic
One numeric column, one time column	Time series
Multiple rows per unit, times as columns	Repeated measures / wide to long
One row per unit, two+ numeric columns	Regression or multivariate
One numeric column, one grouping column	ANOVA
Two categorical columns	Categorical
One column with NAs	Missing data
One column with weird strings	Data cleaning
IDs and measurements	Merging

5 Part 3 — Time Series Decision Tree

5.1 3.1 Overall flow

Q: Is there a time index? - Yes → continue. - No → not a time series question.

Q: What does the question ask?

Sub-question	Go to
Plot / describe	3.2
Stationary?	3.3
Suggest a model	3.4
Fit a model	3.5
Check adequacy	3.6
Forecast	3.7
Missing values	3.8
Compare models	3.9
Seasonal	3.10
Derive theory	3.11

5.2 3.2 “Plot and describe”

What you see	What you say
Upward drift	“Upward trend”
Repeating cycle	“Seasonality with period s”
Amplitude grows with level	“Multiplicative seasonality; consider log transform”
Stable mean and variance	“Stationary-looking”
Sharp spike	“Possible outlier at time t”
Sudden level change	“Structural break”

Code:

```
autoplot(ts_y) + theme_minimal()
ggseasonplot(ts_y)           # if frequency > 1
ggsubseriesplot(ts_y)
```

Output: 2–3 sentence paragraph.

5.3 3.3 “Is the series stationary?”

Step 1: Plot it. Look for trend or changing variance. **Step 2:** Run `adf.test(ts_y)` and `kpss.test(ts_y)`. **Step 3:** If they disagree, look at the ACF.

ADF p	KPSS p	Conclusion
< 0.05	> 0.05	Stationary
> 0.05	< 0.05	Non-stationary; difference
< 0.05	< 0.05	Contradiction; look at plots
> 0.05	> 0.05	Ambiguous; investigate

If non-stationary:

```
dy <- diff(ts_y)
# re-test dy
```

Output: “The series is [stationary / non-stationary] at the 5% level based on the ADF test ($p = \dots$). After first differencing, stationarity is achieved (ADF $p = \dots$).”

5.4 3.4 “Suggest a model”

Step 1: Check stationarity (3.3). **Step 2:** If non-stationary, difference first. **Step 3:** Plot ACF and PACF.

ACF pattern	PACF pattern	Model
Cuts after q	Tails off	MA(q)
Tails off	Cuts after p	AR(p)
Both cut	Both cut	ARMA(p, q)
Both tail off	Both tail off	ARMA(1,1)
Slow decay	Slow decay	ARIMA(p, 1, q)
Spike at lag s	—	Seasonal MA
—	Spike at lag s	Seasonal AR

Code:

```
acf(ts_y, main = "ACF")
pacf(ts_y, main = "PACF")
```

Output: “The ACF cuts off after lag 2; PACF tails off. Suggest MA(2).”

5.5 3.5 “Fit a model”

If they specify order	Do
“Fit AR(1)”	<code>Arima(ts_y, order = c(1, 0, 0))</code>
“Fit MA(2)”	<code>Arima(ts_y, order = c(0, 0, 2))</code>
“Fit ARMA(1,1)”	<code>Arima(ts_y, order = c(1, 0, 1))</code>

If they specify order	Do
“Fit ARIMA(p,d,q)”	<code>Arima(ts_y, order = c(p, d, q))</code>
“Fit the best model”	<code>auto.arima(ts_y)</code>
“Fit a seasonal model”	<code>auto.arima</code> handles it

Output: Model summary + coefficients + AIC.

5.6 3.6 “Is the model adequate?”

Step 1: `checkresiduals(fit)`. **Step 2:** Look at residual ACF and Ljung-Box.

Ljung-Box p	Conclusion
> 0.05	Adequate
< 0.05	Residual structure remains

Also check: - Normality: `shapiro.test(residuals(fit))`. - No visible pattern in residuals over time.

Output: “Ljung-Box p = ...; residuals show no significant autocorrelation. Model is adequate.”

5.7 3.7 “Forecast h steps”

```
fc <- forecast(fit, h = 10)
fc
autoplot(fc) + theme_minimal()
```

Output: - Point forecast at each horizon. - 80% and 95% prediction intervals. - Sentence: “The forecast for time T+1 is X with 95% PI [L, U]. The interval widens with horizon.”

Common trap: Forgetting that intervals widen with h.

5.8 3.8 “Handle missing values”

Situation	Action
Few missing	<code>auto.arima</code> handles them via Kalman filter
Many missing	Impute or use <code>na.interp</code>
Question asks for interpolation	<code>na.interp(ts_y)</code>

Output: “The Kalman filter estimates the likelihood using only observed values. Coefficient estimates are consistent.”

5.9 3.9 “Compare models”

Comparison type	Method
Nested (e.g., AR(1) vs AR(2))	Likelihood ratio test
Non-nested	AIC / BIC
Both	Report both

Output: Table with AIC, BIC, log-likelihood. Say which wins and why.

5.10 3.10 “Seasonal / frequency > 1”

Check: Does the data have `frequency = 12` (monthly) or `frequency = 4` (quarterly)?

Model: `SARIMA(p, d, q)(P, D, Q)s`. - Seasonal difference: `diff(ts_y, lag = s)`. - `auto.arima` picks P, D, Q automatically.

Output: “The fitted model is `SARIMA(1, 1, 1)(1, 1, 1)12`.”

5.11 3.11 Theory questions

Question	Answer approach
“Derive ACF of AR(1)”	Use Yule-Walker; $\rho(k) = \phi^k$
“Show AR(1) is stationary iff...”	Root of $\phi(z) = 0$ outside unit circle
“Write the likelihood of AR(1)”	$y_1 \sim N(\mu, \sigma^2/(1 - \phi^2))$; $y_t y_{t-1}$ normal
“Explain differencing”	Removes stochastic trend; converts random walk to white noise

Cross-reference: Time series cheatsheet Part 22.

6 Part 4 — Regression Decision Tree

6.1 4.1 Overall flow

Q: Is the response continuous, binary, or count? - Continuous $\rightarrow$ `lm()`. - Binary $\rightarrow$ `glm(family = binomial)`. - Count $\rightarrow$ `glm(family = poisson)` or `MASS::glm.nb`. - Time-to-event $\rightarrow$ survival (likely out of scope).

Q: How many predictors? - One $\rightarrow$ simple regression. - Multiple $\rightarrow$ multiple regression.

Q: Any categorical predictors? - Yes $\rightarrow$ dummy variables or factor.

Q: Any interactions? - Yes $\rightarrow$ `y ~ A * B`.

6.2 4.2 By question type

Sub-question	Go to
Fit a model	4.3
Interpret coefficients	4.4
Test if coefficient is zero	4.5
Test multiple coefficients	4.6
Compare models	4.7
Check assumptions	4.8
Predict	4.9
Transform response	4.10
Handle categorical predictors	4.11
Interaction	4.12
Logistic regression	4.13
Poisson regression	4.14

6.3 4.3 “Fit a model”

Simple: `lm(y ~ x, data = df)`. **Multiple:** `lm(y ~ x1 + x2 + x3, data = df)`. **With interaction:** `lm(y ~ x1 * x2, data = df)`. **With factor:** `lm(y ~ factor(group), data = df)`.

Output: `summary(fit)` — coefficients with SE, t, p; R^2 ; F.

6.4 4.4 “Interpret coefficients”

For each coefficient:

Type	Template
Intercept	“Predicted y when all predictors are 0.”
Continuous slope	“A 1-unit increase in x is associated with a -unit change in y, holding others fixed.”
Dummy (level B vs ref A)	“Group B has mean y -units different from reference group A.”
Interaction	“The effect of x1 on y changes by per unit increase in x2.”
Log-transformed y	“A 1-unit increase in x multiplies y by $\exp(\)$.”
Log-log	“A 1% increase in x is associated with a % change in y.”

6.5 4.5 “Is x significant?”

Single coefficient: Look at t-test in `summary(fit)`.

p-value	Conclusion
< 0.05	x is significantly associated with y
0.05	No significant evidence

Report: $t(df) = \dots, p = \dots$

6.6 4.6 “Test multiple coefficients jointly”

Nested model comparison:

```
anova(lm(y ~ x1, data = df), lm(y ~ x1 + x2 + x3, data = df))
```

Linear restriction:

```
car::linearHypothesis(fit, "x1 = x2")
car::linearHypothesis(fit, "x1 + x2 = 5")
```

Output: F-statistic, df, p-value.

6.7 4.7 “Compare models”

Models	Method
Nested	<code>anova(reduced, full)</code>
Non-nested	AIC / BIC
Same df, different predictors	Adjusted R^2
Cross-validation	<code>cv.glm</code> or manual

6.8 4.8 “Check assumptions”

Assumption	Test	Fix
Linearity	Residuals vs fitted	Add polynomial, transform
Normality	Shapiro-Wilk on residuals	Transform, robust SE
Homoskedasticity	Breusch-Pagan	WLS, robust SE
Independence	Durbin-Watson	AR model, GLS
No multicollinearity	VIF	Drop, ridge
No influential points	Cook's D	Report with/without

Output: Paragraph + plots.

6.9 4.9 “Predict for new data”

```
newdf <- data.frame(x = 5)
predict(fit, newdf, interval = "confidence") # mean response
predict(fit, newdf, interval = "prediction") # new observation
```

Report: “Point prediction = X. 95% CI for mean = [L, U]. 95% PI for new obs = [L, U].”

Trap: CI for mean is narrower than PI for a new observation.

6.10 4.10 “Try lambda = -2, -1, 0, 1, 2” (Box-Cox)

Loop over lambdas, compare log-likelihood or R^2 .

Best lambda	Transform
1	None
0	log
0.5	sqrt
-1	1/y

Code:

```
boxCox(lm(y ~ x, data = df))
```

Output: Table of lambda → logLik. Pick the max.

6.11 4.11 “Categorical predictor”

Factors:

```
df$group <- factor(df$group, levels = c("ref", "level1", "level2"))
lm(y ~ group, data = df)
```

Output: Intercept = mean of reference; other coefficients = differences.

If reference changes:

```
df$group <- relevel(df$group, ref = "level1")
```

6.12 4.12 “Interaction”

Question	Model
Does effect of x depend on group?	<code>y ~ x * group</code>
Does effect of x1 depend on x2?	<code>y ~ x1 * x2</code>
Test interaction	<code>anova(y ~ x1 + x2, y ~ x1 * x2)</code>

Interpretation when interaction is significant: Main effects are conditional on the other variable being at its reference/zero.

6.13 4.13 “Logistic regression”

Sub-question	Action
Fit model	<code>glm(y ~ x, family = binomial)</code>
Odds ratio	<code>exp(coef(fit))</code>
CI for OR	<code>exp(confint(fit))</code>
Predict probability	<code>predict(fit, newdata, type = "response")</code>
Compare models	<code>anova(reduced, full, test = "Chisq")</code>
Grouped binomial	<code>glm(cbind(succ, fail) ~ x, family = binomial)</code>

Output: OR + CI + interpretation in terms of odds.

6.14 4.14 “Poisson regression”

Sub-question	Action
Fit model	<code>glm(y ~ x, family = poisson)</code>
Rate ratio	<code>exp(coef(fit))</code>
Overdispersion check	<code>Pearson ^ 2 / df</code>
Fix overdispersion	<code>family = quasipoisson</code> or <code>MASS::glm.nb</code>
Offset (rate model)	<code>glm(y ~ x + offset(log(time)), family = poisson)</code>

Output: Rate ratio + CI + interpretation in terms of counts/rates.

7 Part 5 — Data Handling Decision Tree

7.1 5.1 By task

Sub-task	Go to
Read a file	5.2
Inspect data	5.3
Fix types	5.4
Handle missing values	5.5
Select / rename columns	5.6
Filter / sort rows	5.7
Group summaries	5.8
Reshape wide long	5.9

Sub-task	Go to
Merge / join	5.10
Train / test split	5.11
Outliers	5.12

7.2 5.2 “Read the file”

File type	Command
Excel	<code>read_excel("f.xls", sheet = 1)</code>
CSV	<code>read_csv("f.csv")</code>
SPSS	<code>read_sav("f.sav")</code>
Text	<code>read_table("f.txt")</code>

After reading, always:

```
dim(df); str(df); head(df); colSums(is.na(df))
```

7.3 5.3 “Inspect the data”

Question	Command
How many rows / cols?	<code>dim(df), nrow(df), ncol(df)</code>
Column names?	<code>names(df)</code>
Column types?	<code>str(df), glimpse(df), sapply(df, class)</code>
Sample values?	<code>head(df), tail(df)</code>
Summary stats?	<code>summary(df)</code>
Missing count?	<code>colSums(is.na(df))</code>
Unique values?	<code>unique(df\$col), n_distinct(df\$col)</code>

Output: 1–2 sentence summary.

7.4 5.4 “Fix column types”

Situation	Command
Character → numeric	<code>as.numeric(x)</code>
Character → factor	<code>factor(x, levels = ...)</code>
Factor → character	<code>as.character(x)</code>
Character → Date	<code>as.Date(x, format = ...)</code>
Numeric-coded group	<code>factor(x, levels = ...)</code>

Trap: `as.numeric(factor)` returns level codes, not original values. Use `as.numeric(as.character(f))`.

7.5 5.5 “Handle missing values”

Question	Action
“Remove missing”	<code>drop_na(df)</code> or <code>na.omit(df)</code>

Question	Action
“Replace with mean”	<code>df\$x[is.na(df\$x)] <- mean(df\$x, na.rm = TRUE)</code>
“Use functions with na.rm”	<code>mean(x, na.rm = TRUE)</code>
“Impute”	<code>mice</code> , <code>na.interp</code> , or interpolation
“Report missingness”	<code>colSums(is.na(df)) / nrow(df)</code>

Output: Report fraction missing and approach.

7.6 5.6 “Select / rename columns”

Task	Command
Select by name	<code>select(df, x, y)</code>
Select by pattern	<code>select(df, starts_with("q"))</code>
Drop columns	<code>select(df, -id)</code>
Rename	<code>rename(df, new = old)</code>
Reorder	<code>relocate(df, x, .after = id)</code>

7.7 5.7 “Filter / sort rows”

Task	Command
Filter by condition	<code>filter(df, x > 5)</code>
Filter by multiple	<code>filter(df, x > 5, y < 10)</code>
Filter by membership	<code>filter(df, group %in% c("A", "B"))</code>
Sort ascending	<code>arrange(df, x)</code>
Sort descending	<code>arrange(df, desc(x))</code>

7.8 5.8 “Group summaries”

Task	Command
Mean by group	<code>df %>% group_by(g) %>% summarise(m = mean(x))</code>
Multiple stats	<code>summarise(n = n(), mean = mean(x), sd = sd(x))</code>
Across columns	<code>summarise(across(where(is.numeric), mean))</code>
Count	<code>count(df, g)</code>

7.9 5.9 “Reshape”

Task	Command
Wide → long	<code>pivot_longer(cols, names_to, values_to)</code>
Long → wide	<code>pivot_wider(names_from, values_from)</code>
Separate one column	<code>separate(col, into, sep)</code>
Unite two columns	<code>unite("new", col1, col2, sep)</code>

Output: Reshaped data frame + verify dimensions.

7.10 5.10 “Merge / join”

Task	Command
Keep all from left	<code>left_join(a, b, by = "id")</code>
Keep only matches	<code>inner_join(a, b, by = "id")</code>
Keep all from both	<code>full_join(a, b, by = "id")</code>
Rows in a not in b	<code>anti_join(a, b, by = "id")</code>

After **join**: check `nrow()` before and after.

7.11 5.11 “Train / test split”

```
set.seed(42)
idx <- sample(1:nrow(df), size = 0.8 * nrow(df))
train <- df[idx, ]
test <- df[-idx, ]
```

Report: “Split 80/20, giving X train and Y test rows.”

7.12 5.12 “Detect outliers”

Method	When
IQR rule	Univariate
Z-score	Univariate, normal-ish
Cook’s D	Regression
Mahalanobis	Multivariate

Output: Which rows are flagged + what to do.

8 Part 6 — ANOVA / Experimental Design Decision Tree

8.1 6.1 Overall flow

Q: What is the design?

Design	Go to
3+ independent groups, one factor	6.2
Two factors	6.3
Blocking	6.4
Latin Square	6.5
Split-plot	6.6
Repeated measures	6.7
Mixed (between + within)	6.8
ANCOVA	6.9
Non-normal response	6.10
Unbalanced	6.11
Power / sample size	6.12

8.2 6.2 “One-way ANOVA”

Question forms: - “Are group means equal?” - “Is there a treatment effect?” - “Compare 3+ groups.”

Code:

```
fit <- aov(y ~ group, data = df)
summary(fit)
```

If significant:

```
TukeyHSD(fit)
emmeans::emmeans(fit, ~ group) %>% pairs()
```

Output: F, df, p, post-hoc pairs.

8.3 6.3 “Two-way factorial”

Question forms: - “Does A affect y?” - “Does B affect y?” - “Is there an interaction?”

Code:

```
fit <- aov(y ~ A * B, data = df)
summary(fit)
```

If interaction significant:

```
emmeans::emmeans(fit, ~ A | B) %>% pairs()
```

Output: Three F-tests (A, B, A:B) + simple effects if interaction.

8.4 6.4 “Randomized Complete Block Design (RCBD)”

Question forms: - “Blocks were used. Analyse.” - “Control for block effects.”

Code:

```
fit <- aov(y ~ treatment + block, data = df)
summary(fit)
```

Check additivity:

```
anova(aov(y ~ treatment + block, data = df),
      aov(y ~ treatment * block, data = df))
```

Output: F for treatment and block; additive model assumed.

8.5 6.5 “Latin Square / BIBD”

Question forms: - “Two blocking factors.” - “Each treatment appears once per row/column.”

Code:

```
fit <- aov(y ~ row + col + trt, data = df)
summary(fit)
```

BIBD: more complex; typically reported via `agricolae::BIB.test`.

8.6 6.6 “Split-plot design”

Question forms: - “Hard-to-change factor at whole-plot level.” - “Two error terms.”

Code:

```
fit <- aov(y ~ whole * split + Error(block / whole), data = df)
summary(fit)
```

Read carefully: whole-plot effects in the first stratum; split-plot effects in the second.

8.7 6.7 “Repeated measures”

Question forms: - “Same subjects measured multiple times.” - “Within-subject factor.”

Code:

```
fit <- aov(y ~ time + Error(subject / time), data = df)
summary(fit)
```

Report F-test in subject:time stratum. Check sphericity with `afex::aov_ez`.

Alternative:

```
lmerTest::lmer(y ~ time + (1 | subject), data = df)
```

8.8 6.8 “Mixed design”

Question forms: - “Between-subject group, within-subject time.”

Code:

```
fit <- aov(y ~ group * time + Error(subject / time), data = df)
summary(fit)
```

Output: group in the between-subject stratum; time and interaction in the within-subject stratum.

8.9 6.9 “ANCOVA”

Question forms: - “Compare groups adjusting for a covariate.”

Code:

```
fit <- lm(y ~ group + x, data = df)
car::Anova(fit, type = "III")
emmeans::emmeans(fit, ~ group)
```

Check homogeneity of slopes first:

```
anova(lm(y ~ group + x, data = df),
      lm(y ~ group * x, data = df))
```

8.10 6.10 “Non-normal response”

Test	When
Kruskal-Wallis	One-way, non-normal
Friedman	Repeated measures, non-normal
ART	Factorial, non-normal

Output: ² statistic (KW) or F (ART) + post-hoc.

8.11 6.11 “Unbalanced design”

Use `car::Anova(type = "III")`. Report Type III SS explicitly.

8.12 6.12 “Power / sample size”

Question	Command
Power for one-way	<code>pwr.anova.test(k, n, f, sig.level)</code>
Sample size for power	<code>pwr.anova.test(k, f, power = 0.8)</code>
Power for two-way	<code>pwr.f2.test(u, v, f2)</code>
Power for paired	<code>pwr.t.test(d, type = "paired")</code>

Output: Power or n + interpretation.

9 Part 7 — Multivariate Decision Tree

9.1 7.1 Overall flow

Q: How many response variables? - 1 → not multivariate. - 2+ → continue.

Q: What is the goal?

Goal	Section
Compare mean vectors (2 groups)	7.2
Compare mean vectors (3+ groups)	7.3
Test MVN	7.4
Estimate covariance structure	7.5
Reduce dimensions	7.6
Latent factors	7.7
Classify	7.8
Cluster	7.9
Outliers	7.10
Correlation between two sets	7.11

9.2 7.2 “Two-group Hotelling T^2 ”

Question forms: - “Do the two groups differ on the joint response?” - “Compare mean vectors.”

Code: See multivariate cheatsheet Part 3.2 — manual implementation.

Output: T^2 , F , p + univariate follow-ups.

9.3 7.3 “MANOVA”

Question forms: - “Do 3+ groups differ on the joint response?”

Code:

```
fit <- manova(cbind(y1, y2, y3) ~ group, data = df)
summary(fit, test = "Wilks")
summary.aov(fit) # univariate follow-ups
```

Output: Wilks’ Λ , F , p + follow-up ANOVAs.

9.4 7.4 “Test MVN”

Question forms: - “Is the data multivariate normal?”

Code: Mardia’s test — manual function from multivariate cheatsheet Part 5.3.

Output: Skewness and kurtosis p-values + QQ plot.

9.5 7.5 “Covariance structure”

Question forms: - “Estimate compound symmetry.” - “Check sphericity.”

Code: See multivariate cheatsheet Part 6.2 (CS) and 14.1 (sphericity).

Output: $\hat{\sigma}^2$, $\hat{\rho}$, and interpretation.

9.6 7.6 “PCA”

Question forms: - “Reduce dimension.” - “Find components.” - “Interpret loadings.”

Code:

```
pca <- prcomp(df, scale. = TRUE)
summary(pca)
pca$rotation # loadings
pca$x        # scores
```

Output: 1. Eigenvalues + variance explained. 2. How many components to keep (Kaiser, scree, cumulative). 3. Loadings interpretation. 4. Naming of components.

9.7 7.7 “Factor analysis”

Question forms: - “Find latent factors.” - “Run factor analysis.”

Code:

```
factanal(df, factors = k, rotation = "varimax")
```

Before FA: Bartlett sphericity + KMO. **After FA:** Loadings + factor names + model fit p-value.

9.8 7.8 “Discriminant analysis”

Question forms: - “Classify observations into known groups.”

Code:

```
fit <- MASS::lda(group ~ ., data = df)
predict(fit)
```

Output: Group means, discriminant coefficients, confusion matrix, accuracy.

9.9 7.9 “Cluster analysis”

Question forms: - “Group similar observations.”

Code:

```
hc <- hclust(dist(scale(X)), method = "ward.D2")
plot(hc)
cutree(hc, k = 3)

km <- kmeans(scale(X), centers = 3, nstart = 25)
```

Output: Dendrogram + cluster sizes + centroids + interpretation.

9.10 7.10 “Outliers”

Question forms: - “Detect multivariate outliers.”

Code:

```
md <- mahalanobis(X, colMeans(X), cov(X))
which(md > qchisq(0.975, df = ncol(X)))
```

Output: Which observations exceed the threshold.

9.11 7.11 “Canonical correlation”

Question forms: - “Correlate two sets of variables.”

Code:

```
cancor(X, Y)
```

Output: Canonical correlations + Wilks Λ + interpretation of first pair.

10 Part 8 — Sampling Decision Tree

10.1 8.1 Overall flow

Q: Is there a finite population of known size N ? - Yes $\rightarrow$ continue. - No $\rightarrow$ not a sampling question.

Q: What is the question asking?

Goal	Section
Estimate population mean (SRS)	8.2
Stratified estimate	8.3
Cluster sample	8.4
Systematic sample	8.5
Compare designs	8.6
Bias / variance / MSE	8.7
Coverage probability	8.8
Sample size	8.9
Optimal allocation	8.10
Ratio / regression estimator	8.11
Design effect	8.12

10.2 8.2 “Estimate population mean (SRS)”

Code:

```
s <- sample(pop, n)
ybar <- mean(s)
var_ybar <- (1 - n/N) * var(s) / n
se <- sqrt(var_ybar)
ci <- ybar + c(-1, 1) * qnorm(0.975) * se
```

Output: Point estimate, SE, 95% CI.

10.3 8.3 “Stratified estimate”

Code: See sampling cheatsheet Part 4.3.

Output: $\bar{y}_{st}$, SE, CI + comparison to SRS.

10.4 8.4 “Cluster sample”

Code: See sampling cheatsheet Part 6.4.

Output: Estimator of total, SE, design effect.

10.5 8.5 “Systematic sample”

Code: See sampling cheatsheet Part 5.2.

Output: $\bar{y}_{sys}$, variance (SRS approximation), interpretation.

10.6 8.6 “Compare designs”

Code: Compute variance under each design, take ratio.

Output: Table of variances + relative efficiency + which design wins.

10.7 8.7 “Bias / variance / MSE”

Simulation:

```
estimates <- replicate(B, { ... })
bias <- mean(estimates) - truth
var <- var(estimates)
mse <- mean((estimates - truth)^2)
```

Output: Table of bias, variance, MSE for each estimator.

10.8 8.8 “Coverage”

Simulation:

```
cover <- replicate(B, {
  s <- sample(pop, n)
  ci <- ci_function(s, N, n)
  truth >= ci[1] & truth <= ci[2]
})
mean(cover)
```

Output: Coverage probability + interpretation.

10.9 8.9 “Sample size”

Target	Formula
Variance V	$n = S^2 / (V + S^2 / N)$
Margin of error e	$n = z^2 S^2 / e^2$, adjust with FPC
Proportion	$n = z^2 p(1 - p) / e^2$

Output: Minimal n + rounding up.

10.10 8.10 “Optimal allocation”

Type	Formula
Proportional	$n_h = nN_h/N$
Neyman	$n_h = nN_h S_h / \sum N_j S_j$
Cost-optimal	$n_h = n \frac{N_h S_h / \sqrt{c_h}}{\sum N_j S_j / \sqrt{c_j}}$

Output: Table of n_h + total + variance.

10.11 8.11 “Ratio / regression estimator”

Code: See sampling cheatsheet Part 8.

Output: Point estimate + SE + comparison to SRS.

10.12 8.12 “Design effect”

$$\text{deff} = \text{Var}_{\text{design}} / \text{Var}_{\text{SRS}}$$

Output: deff + effective sample size + interpretation.

11 Part 9 — Categorical Decision Tree

11.1 9.1 Overall flow

Q: How many categorical variables? - One → goodness-of-fit. - Two → independence / homogeneity. - Paired binary → McNemar. - Three → Mantel-Haenszel or log-linear.

Q: What is the sample size?

Expected counts	Test
All 5	Chi-square
Any < 5	Fisher exact
Sparse r×c	Chi-square with simulation

11.2 9.2 By question

Sub-question	Section
Are two variables associated?	9.3
Same distribution across groups?	9.4
Goodness-of-fit	9.5
Small sample	9.6
Before/after paired	9.7
Odds ratio	9.8
Strength of association	9.9
Controlling for a third variable	9.10
Comparing proportions	9.11
Simpson’s paradox	9.12

11.3 9.3 “Are two variables associated?”

Code:

```
tab <- table(df$A, df$B)
chisq.test(tab)
```

Also check: - `chisq.test(tab)$expected` — all 5? - If association significant, `cramers_v(tab)`.

Output: χ^2 , df, p, Cramér’s V.

11.4 9.4 “Same distribution across groups?”

Same computation as 9.3, but report as homogeneity.

11.5 9.5 “Goodness-of-fit”

Code:

```
chisq.test(observed, p = expected_proportions)
```

Output: χ^2 , df = k - 1, p.

11.6 9.6 “Small sample”

Code:

```
fisher.test(tab)
```

Output: Exact p-value + odds ratio + CI.

11.7 9.7 “Before/after paired”

Code:

```
mcnemar.test(tab)
```

If b + c < 25:

```
binom.test(b, b + c, 0.5)
```

Output: χ^2 , df = 1, p.

11.8 9.8 “Odds ratio”

Code:

```
a <- tab[1,1]; b <- tab[1,2]; c <- tab[2,1]; d <- tab[2,2]
OR <- (a*d)/(b*c)
se <- sqrt(1/a + 1/b + 1/c + 1/d)
ci <- exp(log(OR) + c(-1, 1) * qnorm(0.975) * se)
```

Output: OR + 95% CI + interpretation.

11.9 9.9 “Strength of association”

Table	Measure
2×2	Phi, odds ratio
r×c	Cramér’s V

Table	Measure
Ordinal	Kendall , Spearman

Output: Value + benchmark (small/medium/large).

11.10 9.10 “Controlling for a third variable”

Code:

```
mantelhaen.test(array)
```

Output: MH χ^2 , common OR, CI.

11.11 9.11 “Comparing proportions”

Code:

```
prop.test(x = c(x1, x2), n = c(n1, n2))
```

Output: χ^2 , p, CI for difference.

11.12 9.12 “Simpson’s paradox”

Steps: 1. Compute marginal proportions. 2. Compute conditional proportions within strata. 3. If direction reverses, it’s Simpson’s paradox. 4. Adjust with logistic regression or MH.

Output: Marginal vs conditional table + explanation.

12 Part 10 — Cross-Topic Patterns

12.1 10.1 Questions that span two topics

Scenario	Topics	Approach
“Regress y on x, then check residuals”	Regression + Diagnostics	Fit, then 4 plots
“Fit ARIMA, then forecast”	Time series + Forecast	Fit, then forecast
“Design an experiment, then analyze”	Design + ANOVA	Choose design, then aov
“Sample then estimate”	Sampling + Estimation	Draw sample, compute estimator
“Take a contingency table and test”	Categorical + Test	Build table, then chi-square
“Reduce dimensions then classify”	PCA + LDA	PCA scores → LDA

12.2 10.2 Mixed-design questions

Mixed cue	Do
“Controlling for”	Include covariate; ANCOVA or regression
“Adjusting for”	Same
“After removing the effect of”	Residualise or use ANCOVA

Mixed cue	Do
“Given”	Conditioning
“Compare with and without”	Two models or stratified analysis

12.3 10.3 Multi-part questions

Structure: Part (a), (b), (c). **Strategy:** Solve in order; use results of (a) in (b) if natural.

Common patterns:

Sequence	Interpretation
Fit → diagnose → forecast	Standard time-series pipeline
Describe → test → post-hoc	Standard ANOVA pipeline
Fit → compare → predict	Regression pipeline
Read → clean → analyze	Data handling pipeline

13 Part 11 — Red Flags and Traps

13.1 11.1 Words that warn you

Word	What it means
“Controlling for”	Use ANCOVA or include covariate
“After adjusting for”	Same
“Independently”	Test independence, not homogeneity
“Paired”	Use paired test (McNemar, paired t, RM)
“Compare”	Compute a difference and test it
“Interpret”	Give a sentence, not a number
“Explain why”	Not a calculation; explain
“Assume”	You can use that assumption
“Check”	Diagnostic must be run
“Approximately”	CLT applies
“Prove”	Derivation needed
“Show that”	Same
“Verify”	Numerical check

13.2 11.2 Traps by topic

13.2.1 Time series

Trap	Fix
Forgetting to convert to ts	ts(y, frequency = ...) before acf
Wrong frequency	1 annual, 4 quarterly, 12 monthly
Misreading ACF/PACF	ACF cuts → MA; PACF cuts → AR
Forgetting to difference	Test ADF; difference if non-stationary
Comparing models with different d	Illegal

13.2.2 Regression

Trap	Fix
Forgetting to check assumptions	Always plot residuals
Interpreting main effects with interaction	Use simple effects
Using R^2 to compare non-nested models	Use AIC/BIC
Confusing CI and PI	CI for mean; PI for new obs
Ignoring multicollinearity	Check VIF

13.2.3 ANOVA

Trap	Fix
Using <code>ao</code> without <code>Error</code> for RM	Use <code>Error(subject/time)</code>
Reading top stratum in RM	Read within-subject stratum
Type I SS on unbalanced	Use Type III
Using Tukey for unequal variances	Use Games-Howell
Multiple tests without correction	Bonferroni, Tukey, or FDR

13.2.4 Multivariate

Trap	Fix
Not standardizing before PCA	Use <code>scale. = TRUE</code>
Confusing <code>sdev</code> with eigenvalues	Square <code>sdev</code>
Reading loadings wrong	Rows = variables, cols = components
Treating PCA as FA	PCA = components, FA = latent factors
Using k-means with non-spherical clusters	Use hierarchical

13.2.5 Sampling

Trap	Fix
Forgetting FPC	Multiply variance by $(1 - n/N)$
Confusing N and n	N = population, n = sample
Using SRS variance for cluster	Apply design effect
Not setting seed	<code>set.seed()</code> before <code>sample</code>

13.2.6 Categorical

Trap	Fix
Chi-square with expected < 5	Use Fisher
Confusing independence and homogeneity	Same test, different design
Using McNemar on unpaired data	Check pairing
Reporting OR without CI	Always give CI
Ignoring Simpson's paradox	Stratify before concluding

14 Part 12 — Output Shape Templates

14.1 12.1 Estimate

“The point estimate is X ($SE = Y$). A 95% confidence interval is $[L, U]$.”

14.2 12.2 Test

“The [test name] gives statistic = value, $p =$ value. [Reject / Fail to reject] H_0 at the 5% level. There is [significant / no significant] evidence that [conclusion].”

14.3 12.3 Model fit

“The fitted model is [equation]. The coefficient of x is (SE = s , $p = p$). $R^2 = \dots$. The model explains $X\%$ of the variance.”

14.4 12.4 Interpretation

“A 1-unit increase in x is associated with a -unit [increase / decrease] in y , holding other predictors constant.”

14.5 12.5 Forecast

“The forecast for the next h steps is [values]. The 95% prediction interval is $[L, U]$. The interval widens with horizon.”

14.6 12.6 Post-hoc

“[Method] shows that groups [A and B] differ significantly (mean difference = Δ , 95% CI $[L, U]$, $p = p$).”

14.7 12.7 Diagnostics

“Shapiro-Wilk on residuals gives $p = X$. Levene’s test gives $p = Y$. Q-Q plot shows points on the diagonal. Assumptions are satisfied.”

14.8 12.8 Effect size

“[Measure] = value, indicating a [small / medium / large] effect.”

14.9 12.9 Comparison

“Model A has $AIC = X$, Model B has $AIC = Y$. Model [A/B] is preferred.”

14.10 12.10 Conclusion

“In summary, [one-sentence answer to the question], with [caveat about assumptions / limitations].”

15 Part 13 — Time Management

15.1 13.1 The 4-3-2-1 rule

For a question worth T minutes:

Phase	Time	Action
Read	10% of T	Underline verb, circle nouns, identify design
Setup	30% of T	Load data, clean, build the required object
Analysis	40% of T	Fit the model, run the test, do post-hoc
Write-up	20% of T	Interpret, conclude, check output shape

15.2 13.2 Decision shortcuts

If you're stuck...	Do this
Can't identify the design	List the variables and their types
Can't pick the test	Match the question shape to Part 0.3
Can't interpret the output	Use the templates in Part 12
Can't remember R syntax	Check the topic cheatsheet for that section
Can't remember a formula	Derive it from the definition
Can't tell if the answer is right	Sanity-check: sign, magnitude, p-value range

15.3 13.3 Common pitfalls in exam timing

1. **Spending 20 minutes on the setup** because the data is messy → clean first, analyse later.
2. **Not checking assumptions** because you're rushing → 3 minutes for diagnostics.
3. **Writing too much prose** → 4-5 sentences per part.
4. **Recomputing the same thing** → save intermediate results.
5. **Forgetting to report the answer** → always end with a sentence.

16 Part 14 — The Master Checklist

Before submitting any question, check:

- ☐ Did you **answer the verb**? (Fit? Test? Estimate? Forecast?)
- ☐ Did you **use the right method** for the design?
- ☐ Did you **check assumptions**?
- ☐ Did you **report the statistic, df, p, and effect size**?
- ☐ Did you **interpret in context**?
- ☐ Did you **mention limitations**?
- ☐ Did you **follow the output shape** (Part 12)?
- ☐ Did you **not skip parts**?

If yes to all eight, you're done.

17 Part 15 — Cross-Reference Table

Where to find each topic's full treatment.

Topic	Cheatsheet	Section
R basics	R Basics	All
Data handling	Data Handling	All
Regression	Regression	All

Topic	Cheatsheet	Section
ANOVA & Design	ANOVA & Experimental Design	All
Time series	Time Series	All
Multivariate	Multivariate	All
Sampling	Sampling	All
Categorical	Categorical	All

Each cheatsheet is self-contained. Every chunk in them runs in isolation.

18 Part 16 — The One Rule

If you can find your question in this guide, you can solve it.

If you can't find it:

1. **Re-read the question.** You probably missed a keyword.
2. **Check Part 10 (Cross-Topic).** It might be a mixed question.
3. **Check Part 11 (Traps).** It might be worded to mislead.
4. **Fall back on Part 0.2.** The universal fallback table.

If after all four steps you still can't place the question, it's out of scope for this exam.

End of question analysis guide. Good luck in the competition. ““

18.1 What this guide covers

Part	Content
0	Meta-strategy: how to read any question, universal fallback, five question shapes
1	Keyword index: 80+ keywords mapped to topics
2	Question shape classifier by verb, structure, and data shape
3	Time series decision tree with 11 sub-branches
4	Regression decision tree with 14 sub-branches
5	Data handling decision tree with 12 sub-branches
6	ANOVA/Design decision tree with 12 sub-branches
7	Multivariate decision tree with 11 sub-branches
8	Sampling decision tree with 12 sub-branches
9	Categorical decision tree with 12 sub-branches
10	Cross-topic patterns and multi-part question strategies
11	Red flags and traps by topic
12	Output shape templates (10 templates)
13	Time management: 4-3-2-1 rule and pitfalls
14	Master checklist (8 items)
15	Cross-reference table
16	The one rule (fallback protocol)

How to use: search for the **verb** (Part 2) or a **keyword** (Part 1). Follow the branch to a “do this”. Run the code from the matching cheatsheet section. Output in the shape from Part 12.